Implementation Walkthrough

This project follows a standard Machine Learning pipeline to predict housing prices. Below is a detailed breakdown of each phase:

1. Data Acquisition & Setup

- **Library Imports**: We use pandas and numpy for data manipulation, matplotlib and seaborn for visualization, and scikit-learn for the machine learning algorithms.
- **Dataset**: The **California Housing Dataset** is a classic regression dataset where the goal is to predict the MedHouseVal (Median House Value) based on 8 features like median income, house age, and location (latitude/longitude).

2. Preprocessing & Feature Engineering

- **Train-Test Split**: We split the data into 80% training and 20% testing sets. This ensures we evaluate our model on unseen data to check for overfitting.
- **Feature Scaling**: Linear models are sensitive to the scale of input features. We apply StandardScaler to normalize the data so that every feature has a mean of 0 and a standard deviation of 1. This prevents features with large values (like population) from dominating those with small values (like bedroom count).

3. Model Training & Selection

We implement two different types of regression algorithms to compare their logic:

- **Linear Regression**: A parametric approach that assumes a linear relationship between features and the target. It is highly interpretable but may struggle with complex, non-linear patterns.
- **Decision Tree Regressor**: A non-parametric approach that splits the data into branches based on feature thresholds. It is excellent at capturing non-linear relationships and interactions between features.

4. Evaluation Metrics

- **RMSE (Root Mean Squared Error)**: Measures the average magnitude of the error. Since it is in the same units as the target ($100,000s), it provides a clear sense of how far off our predictions are.
- **R^2 (Coefficient of Determination)**: Represents the proportion of variance for the target variable that's explained by the features. A score closer to 1.0 indicates a better fit.

5. Visualization

- The **Actual vs. Predicted** scatter plots are the ultimate 'sanity check'. In a perfect model, all points would lie on the red diagonal line. Deviations from this line visualize where the model is underestimating or overestimating prices.